



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (CYBERSECURITY)
SECURE REMOTE NETWORK ACCESS USING TWIN GATE AND RASPBERRY PI

ABSTRACT

The proposed idea focuses on developing a secure, efficient, and cost-effective solution for remote network access by using a Raspberry Pi as a connector for the Twingate Zero Trust Network Access (ZTNA) platform. Traditional remote access methods such as Virtual Private Networks (VPNs) often require firewall modifications, port forwarding, public IP addresses, and complex network configurations. The proposed system addresses these limitations by implementing a zero-trust approach based on identity verification and least-privilege access, ensuring that users can access only the resources for which they are authorized.

The Raspberry Pi functions as a lightweight and affordable network connector that securely links remote devices and services to the Twingate platform. Through its cloud-managed architecture, Twingate simplifies deployment, centralized management, authentication, and access control while reducing the need for exposing internal network services directly to the internet. Security is further enhanced through granular access policies, user authentication, device verification, and activity logging.

To improve scalability and minimize manual configuration, the system integrates the Twingate API with a Python-based automation script. The automation component can discover available network resources and dynamically register them for secure remote access. Additional features, including DNS aliasing, temporary access permissions, and service accounts, support remote administration, network monitoring, maintenance, and authorized security assessments.

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